

CIE-1 PYQ Bank: Model Answers

Handwritten Question Bank & Solutions

Syllabus Topics Covered:

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Unit 1 Examination Questions & Answers

UNIT 1

Verified model solutions for Unit 1 past exam questions.

Unit 1 6 Marks High Priority

SEE 2025SEE JAN 2026

Theory Link: Unit 1 § 1

Define Data Communication and discuss its fundamental characteristics. With a neat diagram illustrate the five components of data communication.

Practice Problem:

Model Exam Answer

Solution:

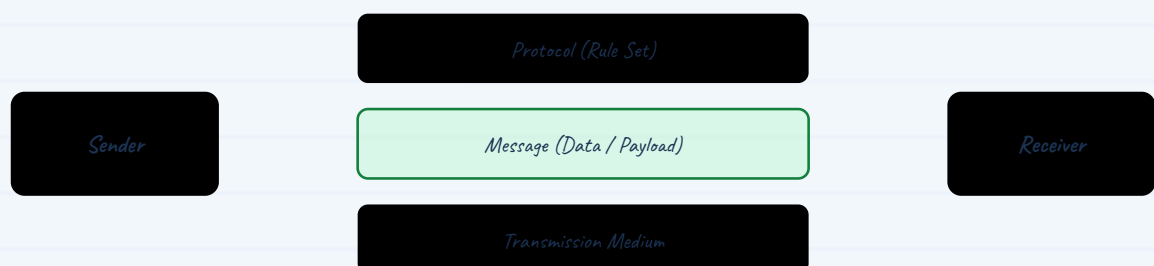
1. Definition of Data Communication

Data communication is the exchange of data between two devices via some form of transmission medium (such as a wire cable or wireless link). For data communication to occur, communicating devices must be part of a communication system made up of a combination of hardware and software.

2. Four Fundamental Characteristics

- **Delivery:** The system must deliver data to the correct destination. Data must be received only by the intended device or user.
- **Accuracy:** The system must deliver the data accurately. Altered data that are left uncorrected are unusable.
- **Timeliness:** The system must deliver data in a timely manner (real-time data delivery without significant delay). For audio and video, late data is useless.
- **Jitter:** Jitter refers to the variation in packet arrival time. It is the uneven delay in the delivery of audio or video packets that can cause jerky reception.

3. Five Components of Data Communication



1. **Message:** The information (data) to be communicated (text, numbers, pictures, audio, video).

2. **Sender:** The device that creates and sends the message (computer,

Parameter	Mesh Topology	Star Topology
Link Architecture	Dedicated point-to-point links between every pair of nodes: $L = \frac{N(N-1)}{2}$ links.	Each node has a dedicated point-to-point link to a central controller (Hub/Switch): N links.
Port Requirement	Every node requires $(N - 1)$ I/O ports.	Each node requires only 1 I/O port; central device needs N ports.
Fault Tolerance	Extremely robust. Failure of one link does not affect any other node communication.	Node link failure isolates only that node. However, central switch failure disables the entire network.
Cost & Installation	High cabling cost and complex installation due to bulk cabling.	Far less expensive and simple to install, configure, and reconfigure.

Layer	Address Type	Size / Format	Example
Application	Specific / Name Address	Variable URL/email string	user@msrit.edu, www.ietf.org
Transport	Port Address	16-bit integer (0 to 65535)	Port 80 (HTTP), Port 443 (HTTPS), Port 53 (DNS)
Network	Logical (IP) Address	32-bit (IPv4) / 128-bit (IPv6)	192.168.1.1 (IPv4)
Data Link	Physical (MAC) Address	48-bit (6 bytes hex)	00:1A:2B:3C:4D:5E

Unit 1 7 Marks High Priority

CIE-1 2025SEE 2024SEE JAN 2026

Theory Link: Unit 1 § 4What is transmission impairment? Briefly explain its causes with necessary diagrams.Practice Problem:

Model Exam Answer

Solution:**1. Definition of Transmission Impairment**

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not identical to the signal at the end of the medium (what is sent is not what is received).

2. Three Causes of Transmission Impairment

1. **Attenuation (Loss of Energy):** When a signal travels through a medium, it loses energy overcoming the resistance of the medium. To compensate for this loss, amplifiers are used to amplify the signal.

Measurement in Decibels: $\text{dB} = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$ where P_1 is input power and P_2 is output power.



2. **Distortion (Alteration of Waveform):** Occurs in composite signals made of different frequencies. Each frequency component has its own propagation speed through a medium and therefore arrives with a different phase delay at the receiver. Consequently, the components combine into a distorted shape.

3. **Noise:** External electrical signals corrupting the original signal. Types include:

- Thermal Noise: Random motion of electrons in a conductor.
- Induced Noise: Comes from sources like motors and appliances (transmitters act as sending antennas and media act as receiving antennas)

Unit 1 4 Marks High Priority

CIE-1 2025

Theory Link: Unit 1 § 5

We have a channel with a 2-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level? (To calculate signal level, consider the bit rate to be 4 Mbps less than the upper limit).

Practice Problem:

Model Exam Answer

Solution:*Step-by-Step Numerical Solution***Given Parameters:**

- Bandwidth $B = 2 \text{ MHz} = 2 \times 10^6 \text{ Hz}$
- Signal-to-Noise Ratio $\text{SNR} = 63$

Step 1: Calculate Theoretical Upper Limit (Shannon Capacity)

By Shannon's Capacity Formula:

$$\begin{aligned}\text{Capacity } C &= B \log_2(1 + \text{SNR}) \\ C &= (2 \times 10^6) \times \log_2(1 + 63) = (2 \times 10^6) \times \log_2(64) \\ \text{Since } 64 &= 2^6 \implies \log_2(64) = 6 \\ C &= 2 \times 10^6 \times 6 = 12 \times 10^6 \text{ bps} = 12 \text{ Mbps}\end{aligned}$$

The upper limit of data rate is **12 Mbps**.*Step 2: Determine Operating Bit Rate*

According to the problem statement, the appropriate bit rate must be 4 Mbps less than the upper limit:

$$\text{Bit Rate } R = C - 4 \text{ Mbps} = 12 \text{ Mbps} - 4 \text{ Mbps} = 8 \text{ Mbps} = 8 \times 10^6 \text{ bps}$$

Step 3: Determine Required Signal Level (M) using Nyquist Bit Rate Formula

By Nyquist's Bit Rate Theorem for a noiseless channel operating at this rate:

$$\begin{aligned}R &= 2 \times B \times \log_2(M) \\ 8 \times 10^6 &= 2 \times (2 \times 10^6) \times \log_2(M) \\ 8 \times 10^6 &= 4 \times 10^6 \times \log_2(M) \\ \log_2(M) &= \frac{8 \times 10^6}{4 \times 10^6} = 2\end{aligned}$$

Unit 1 4 Marks High Priority

CIE-1 2025SEE JAN 2026

Theory Link: Unit 1 § 6

Convert the digital data 11001010 to a digital signal by applying AMI and Differential Manchester line coding schemes.

Practice Problem:

Model Exam Answer

Solution:

Bit Sequence: 1 1 0 0 1 0 1 0

1. Alternate Mark Inversion (AMI) Scheme

- **Rule:** Binary 0 is represented by zero voltage (0 V). Binary 1s are represented by alternating positive (+V) and negative (−V) voltages.
- **Bit-by-Bit Encoding:**
 - 1st bit = 1 → +V
 - 2nd bit = 1 → −V (alternate)
 - 3rd bit = 0 → 0 V
 - 4th bit = 0 → 0 V
 - 5th bit = 1 → +V (alternate)
 - 6th bit = 0 → 0 V
 - 7th bit = 1 → −V (alternate)
 - 8th bit = 0 → 0 V
- **Waveform Levels:** +V, −V, 0, 0, +V, 0, −V, 0

2. Differential Manchester Scheme

- **Rule:** Always contains a mid-bit transition for synchronization. Bit 0 forces a transition at the beginning of the bit interval. Bit 1 has NO transition at the beginning of the bit interval.
- Assuming initial state begins from High voltage:
 - Bit 1 (1): No transition at start; mid-bit transition High → Low.
 - Bit 2 (1): No transition at start; mid-bit transition Low → High.
 - Bit 3 (0): Transition at start (High → Low); mid-bit transition Low → High.
 - Bit 4 (0): Transition at start (High → Low); mid-bit transition Low

Unit 2 Examination Questions & Answers

UNIT 2

Verified model solutions for Unit 2 past exam questions.

Parameter	Byte Stuffing	Bit Stuffing
Orientation	Byte/Character-oriented (e.g., PPP)	Bit-oriented protocols (e.g., HDLC)
Stuffing Unit	Inserts an entire 8-bit byte (ESC)	Inserts a single bit (0)
Overhead	8 bits added per stuffed instance	Only 1 bit added per stuffed instance

Unit 2 7 Marks High Priority

CIE-1 2025SEE 2025

Theory Link: Unit 2 § 2

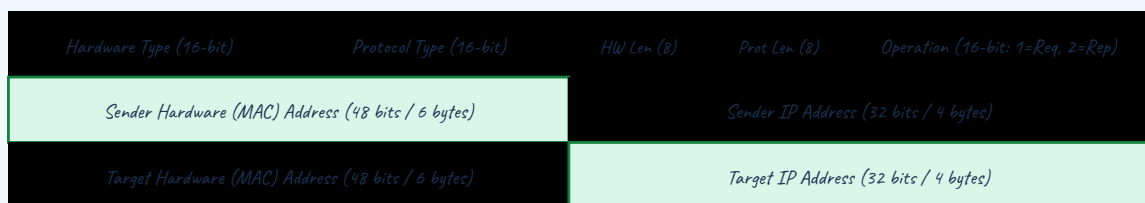
With a neat diagram, discuss the different fields of an ARP packet. Explain the process of ARP in resolving IP addresses to physical MAC addresses.

Practice Problem:

Model Exam Answer

Solution:

1. ARP Packet Format (28 Bytes)



2. Field Explanations:

- **Hardware Type (16-bit):** Defines physical network type (e.g., Ethernet = 1).
- **Protocol Type (16-bit):** Defines protocol used at network layer (IPv4 = 0x0800).
- **Hardware Length (8-bit):** Length of MAC address in bytes (Ethernet = 6).
- **Protocol Length (8-bit):** Length of logical IP address in bytes (IPv4 = 4).
- **Operation Code (16-bit):** 1 for ARP Request, 2 for ARP Reply.
- **Sender Hardware & Protocol Address:** MAC and IP address of the inquiring device.
- **Target Hardware & Protocol Address:** In an ARP Request, Target MAC is unknown (all 0s); Target IP is the destination address to be resolved.

3. ARP Resolution Process:

1. **ARP Request (Broadcast):** Host A wants to send an IP datagram to Host B on the same LAN but knows only Host B's IP address. Host A encapsulates an ARP packet inside a broadcast Ethernet frame with destination MAC FF·FF·FF·FF·FF·FF. Every node on the LAN receives

Unit 2 7 Marks High Priority

CIE-1 2025SEE JAN 2026SEE 2024SEE 2025

Theory Link: Unit 2 § 4

Define cyclic codes and linear block codes. Given the dataword 101011 and the divisor 10011, generate the codeword using CRC polynomial division.

Practice Problem:

Model Exam Answer

Solution:

1. Definitions

- **Linear Block Codes:** A code is linear if the modulo-2 sum (XOR) of any two valid codewords is also a valid codeword in the code. Block codes divide message data into fixed k -bit datawords and append $(n - k)$ parity bits to produce n -bit codewords.
- **Cyclic Codes:** A special subset of linear block codes where cyclically shifting any valid codeword produces another valid codeword. Implemented efficiently in hardware using feedback shift registers and modulo-2 polynomial division.

2. CRC Generation Problem

Given:

- Dataword $D = 101011$ ($k = 6$ bits)
- Divisor $G = 10011$ ($n - k + 1 = 5$ bits $\implies$ polynomial degree $r = 4$)

Step 1: Append $r = 4$ zeros to the dataword

Augmented Dataword = 1010110000
(10 bits total)

Step 2: Modulo-2 Polynomial Division (XOR Division)

```

Dividend: 1010110000 | Divisor: 10011
1010110000 XOR 1001100000 → 0011010000
0011010000 XOR 0010011000 → 0001001000
0001001000 XOR 0001001100 → 0000000100
Final Remainder (4 bits) = 0011

```

Quotient = 101101, Remainder $R = 0011$.

Step 3: Construct Codeword

Unit 2 4 Marks High Priority

CIE-1 2025SEE JAN 2026

Theory Link: Unit 2 § 5

Calculate the Internet checksum for the text 'ISE DEPARTMENT' at sender and receiver side (assume hexadecimal ASCII: I=49, S=53, E=45, space=32, D=44, P=50, A=41, R=52, T=54, M=4D, N=4E).

Practice Problem:

Model Exam Answer

Solution:

1. ASCII to 16-Bit Word Mapping

The text is 14 characters: "ISE DEPARTMENT"

1. "IS" = 4953 (hex)
2. "E " = 4520 (hex) (E = 45, space = 32 in hex = 20)
3. "DE" = 4445 (hex)
4. "PA" = 5041 (hex)
5. "RT" = 5254 (hex)
6. "ME" = 4D45 (hex)
7. "NT" = 4E54 (hex)

2. Sender Side 16-Bit One's Complement Addition

$$4953 + 4520 = 8E73$$

$$8E73 + 4445 = D2B8$$

$$D2B8 + 5041 = 122F9 \rightarrow (22F9 + 1) = 22FA \text{ (wrap carry)}$$

$$22FA + 5254 = 754E$$

$$754E + 4D45 = C293$$

$$C293 + 4E54 = 110E7 \rightarrow (10E7 + 1) = 10E8 \text{ (wrap carry)}$$

$$\text{Sum} = 10E8_{16}$$

$$\text{Checksum (One's complement of Sum)} = \sim (10E8) = EF17_{16}$$

The sender transmits the data words along with the Checksum EF17.

3. Receiver Side Verification

The receiver sums all 7 data words plus the received checksum:

$$\text{Sum of data words} = 10E8_{16}$$

Unit 2 6 Marks High Priority

CIE-2 2025SEE 2024

Theory Link: Unit 2 § 6

Briefly discuss the working of Stop-and-Wait protocol for a noisy channel (Stop-and-Wait ARQ) with necessary flow diagrams.

Practice Problem:

Model Exam Answer

Solution:

1. Working of Stop-and-Wait ARQ

Stop-and-Wait Automatic Repeat reQuest (ARQ) adds error control and flow control mechanisms to simple Stop-and-Wait over noisy channels:

- **Sequence Numbers:** Frames alternate between sequence number 0 and 1 (using 1-bit sequence number $m = 1, 2^1 = 2$).
- **Acknowledgment Numbers:** ACKs announce the sequence number of the next expected frame (ACK 0 or ACK 1).
- **Retransmission Timer:** The sender maintains a timer. If an ACK does not arrive before timer expiration (due to a lost frame or lost ACK), the sender retransmits the copy kept in its buffer.

2. Operation Scenarios:

1. **Normal Operation:** Sender sends Frame 0 → Receiver receives Frame 0, sends ACK 1 → Sender sends Frame 1.
2. **Lost Frame:** Sender sends Frame 0 → Frame lost in transit → Sender timer expires → Sender retransmits Frame 0 → Receiver receives Frame 0, sends ACK 1.
3. **Lost ACK:** Sender sends Frame 0 → Receiver sends ACK 1, but ACK is lost → Sender timer expires → Sender retransmits Frame 0 → Receiver discards duplicate Frame 0 and re-sends ACK 1!

Source: Forouzan Se, Chapter 11.2, pp. 317–324

Unit 2 6 Marks High Priority

CIE-2 2025

Theory Link: Unit 2 § 6

Consider $m=2$, and with necessary diagrams justify the statement: 'Size of the sender window must be equal to 2^{m-1} in Selective Repeat ARQ Protocol'.

Practice Problem:

Model Exam Answer

Solution:

1. Theoretical Principle

In Selective Repeat ARQ, both sender window size W_s and receiver window size W_r must be equal, and satisfy:

$$W_s = W_r = 2^{m-1}$$

For $m = 2$ bits, sequence numbers are 0, 1, 2, 3. The allowable window size is $W = 2^{2-1} = 2$.

2. Proof by Contradiction: What happens if $W > 2^{m-1}$ (e.g., $W = 3$)?

1. Suppose $W = 3$ with sequence numbers 0, 1, 2, 3. Sender window contains frames [0, 1, 2].
2. Sender transmits frames 0, 1, and 2.
3. Receiver correctly receives frames 0, 1, 2 and advances its window to [3, 0, 1].
4. Receiver sends ACKs 0, 1, 2 back to the sender, but **all ACKs are lost in transit!**
5. Sender's timer expires for Frame 0. Sender retransmits Frame 0.
6. The receiver is currently waiting for window [3, 0, 1]. Because 0 is in the receiver's new window, the receiver **mistakes the retransmitted duplicate Frame 0 for the brand new Frame 0 of the next cycle!**
7. This causes fatal undetected duplicate data acceptance.

3. Correct Case when $W = 2^{m-1} = 2$:

With $W = 2$, sender window is [0, 1] and receiver window is [0, 1]. When receiver accepts frames 0 and 1, receiver window slides to [2, 3]. If all ACKs are lost and sender retransmits Frame 0, 0 is completely outside receiver's window [2, 3]. Receiver recognizes it as a duplicate and discards it! Hence,

Unit 2 6 Marks Medium Priority

SEE JAN 2026

Theory Link: Unit 2 § 6

Justify the statement 'Size of the send window must be less than 2^m in Go-Back-N ARQ protocol' with suitable diagrams.

Practice Problem:

Model Exam Answer

Solution:

1. Principle of Go-Back-N Window Sizing

In Go-Back-N ARQ with an m -bit sequence number (2^m sequence numbers 0 to $2^m - 1$), the maximum send window size must satisfy:

$$W_s \leq 2^m - 1$$

2. Justification (What if $W_s = 2^m$?):

- Let $m = 2$, so sequence numbers are 0, 1, 2, 3 ($2^m = 4$). Suppose $W_s = 4$.
- Sender transmits frames 0, 1, 2, 3.
- Receiver receives all 4 frames in order and sends cumulative ACK 4 (meaning expecting Frame 0 of next cycle).
- If ACK 4 is lost, sender times out and retransmits frames 0, 1, 2, 3.
- The receiver is expecting Frame 0 of the next cycle. It cannot distinguish whether Frame 0 is a retransmitted duplicate or new data!
- By limiting $W_s \leq 2^m - 1 = 3$, the window covers at most $[0, 1, 2]$. When ACK is lost, retransmitted Frame 0 can never overlap with next cycle's expected frame.

Source: Forouzan Se, Chapter 11.2, pp. 325-330

Unit 2 6 Marks Medium Priority

SEE 2024

Theory Link: Unit 2 § 6

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system produces: (i) 1000 frames/sec, (ii) 500 frames/sec, (iii) 250 frames/sec?

Practice Problem:

Model Exam Answer

Solution:

1. Parameter Calculations

- Frame length $L = 200$ bits
- Channel bandwidth $R = 200$ kbps $= 200,000$ bps
- Frame transmission time $T_{fr} = \frac{L}{R} = \frac{200}{200,000} = 1 \text{ ms} = 0.001 \text{ sec}$

Throughput formula for Pure ALOHA: $S = Ge^{-2G}$, where $G = \text{load} = N \times T_{fr}$

2. Calculations for Each Load:

1. **Case (i): 1000 frames/sec**

$$G = 1000 \times 0.001 = 1.0$$

$$S = 1.0 \times e^{-2(1.0)} = 1.0 \times e^{-2} = 1.0 \times 0.1353 = 0.1353$$

(13.53%)

$$\text{Throughput in bps} = S \times R = 0.1353 \times 200 \text{ kbps} = 27.06 \text{ kbps}$$

(or 135.3 frames/sec).

2. **Case (ii): 500 frames/sec**

$$G = 500 \times 0.001 = 0.5$$

$$S = 0.5 \times e^{-2(0.5)} = 0.5 \times e^{-1} = 0.5 \times 0.3679 = 0.1839$$

(18.39% — peak capacity!)

$$\text{Throughput in bps} = 0.1839 \times 200 \text{ kbps} = 36.79 \text{ kbps (or 183.9 frames/sec).}$$

3. **Case (iii): 250 frames/sec**

$$G = 250 \times 0.001 = 0.25$$

$$S = 0.25 \times e^{-2(0.25)} = 0.25 \times e^{-0.5} = 0.25 \times 0.6065 = 0.1516$$

(15.16%)

$$\text{Throughput in bps} = 0.1516 \times 200 \text{ kbps} = 30.33 \text{ kbps (or 151.6 frames/sec).}$$

Unit 3 Examination Questions & Answers

UNIT 3

Verified model solutions for Unit 3 past exam questions.

Feature	Datagram Approach (Connectionless)	Virtual-Circuit Approach (Connection-Oriented)
Connection Setup	No setup phase. Packets are injected into the network immediately.	Mandatory 3-phase process: Connection setup, Data transfer, Teardown.
Addressing in Packet	Each packet contains full source and destination IP addresses.	Packets carry a short Virtual Circuit Identifier (VCI / Label).
Routing & Path	Each packet is routed independently. Packets may follow different routes.	All packets follow the exact same predetermined path.
Packet Order	Packets may arrive out of order at destination.	Packets arrive strictly in the order they were transmitted.
Router State	Routers do not hold state information about active connections.	Each intermediate switch maintains a state/forwarding table with active VCIs.
Effect of Switch Failure	Highly robust; remaining packets dynamically reroute around failed switches.	All virtual circuits traversing the failed switch terminate.

Unit 3 4 Marks High Priority

SEE 2025(0)

Theory Link: Unit 3 § 3

What is the purpose of a subnet mask in IPv4 addressing? How is it used to determine the network and host portion of an IP address?

Practice Problem:

Model Exam Answer

Solution:

1. Purpose of a Subnet Mask

A subnet mask is a 32-bit binary number that demarcates which part of an IPv4 address corresponds to the **network prefix** and which part corresponds to the **host identifier**. In dotted-decimal notation, 1s indicate network bits and 0s indicate host bits (e.g., 255.255.255.0 has 24 ones and 8 zeros, written as /24).

2. Bitwise Operation to Extract Network Address

To determine the network address, the router performs a bitwise logical **AND** operation between the destination IP address and the subnet mask:

Network Address = IP Address AND Subnet Mask

- **Example:** Given IP address 192.168.10.75 and mask

255.255.255.192 (/26):

- IP in binary: 11000000.10101000.00001010.01001011
- Mask in binary: 11000000.11111111.11111111.11000000
- Bitwise AND: 11000000.10101000.00001010.01000000 $\Rightarrow$
192.168.10.64

- The remaining 6 bits identify host number $11_2 = 11$ within the subnet 192.168.10.64/26.

Source: Forouzan Se, Chapter 18.4, pp. 531–535

Unit 3 10 Marks High Priority

SEE 2025

Theory Link: Unit 3 § 3

An ISP is granted a block of addresses starting with 120.50.0.0/16. The ISP wants to distribute these addresses to three groups of customers: (i) Group 1: 64 customers, each needing 256 addresses; (ii) Group 2: 128 customers, each needing 128 addresses; (iii) Group 3: 128 customers, each needing 64 addresses. Design the sub-blocks, find first and last address for each customer, and calculate the number of unallocated addresses remaining.

Practice Problem:

Model Exam Answer

Solution:

1. Address Space Analysis

- Total addresses in granted block $120.50.0.0/16 = 2^{32-16} = 2^{16} = 65,536$ addresses.

2. Sub-Block Design per Group

Group 1: 64 customers, each needing 256 addresses

- Each customer needs $256 = 2^8$ addresses $\implies$ host bits = 8 $\implies$ prefix length $n = 32 - 8 = /24$.
- Total addresses for Group 1 = $64 \times 256 = 16,384$ addresses.
- **Customer 1:** 120.50.0.0/24 to 120.50.0.255/24
- **Customer 2:** 120.50.1.0/24 to 120.50.1.255/24
- ...
- **Customer 64:** 120.50.63.0/24 to 120.50.63.255/24
- Group 1 ends at: 120.50.63.255.

Group 2: 128 customers, each needing 128 addresses

- Each customer needs $128 = 2^7$ addresses $\implies$ host bits = 7 $\implies$ prefix length $n = 32 - 7 = /25$.
- Total addresses for Group 2 = $128 \times 128 = 16,384$ addresses.
- Starts at next available IP: 120.50.64.0/25
- **Customer 1:** 120.50.64.0/25 to 120.50.64.127/25
- **Customer 2:** 120.50.64.128/25 to 120.50.64.255/25
- **Customer 3:** 120.50.65.0/25 to 120.50.65.127/25

Unit 3 10 Marks High Priority

SEE JAN 2023

Theory Link: Unit 3 § 3

A large number of consecutive IP addresses are available starting at 198.16.0.0/16. Suppose that four organizations require: Org 1: 4000 addresses, Org 2: 2000 addresses, Org 3: 4000 addresses, Org 4: 8000 addresses. Determine address allocations and subnet masks.

Practice Problem:

Model Exam Answer

Solution:

1. Block Size Rounding (Powers of 2)

1. **Org 4 (8000 addresses):** Smallest power of 2 is $2^{13} = 8192$. Prefix length = $32 - 13 = /19$. Mask: 255.255.224.0.
2. **Org 1 (4000 addresses):** Smallest power of 2 is $2^{12} = 4096$. Prefix length = $32 - 12 = /20$. Mask: 255.255.240.0.
3. **Org 3 (4000 addresses):** Smallest power of 2 is $2^{12} = 4096$. Prefix length = $32 - 12 = /20$. Mask: 255.255.240.0.
4. **Org 2 (2000 addresses):** Smallest power of 2 is $2^{11} = 2048$. Prefix length = $32 - 11 = /21$. Mask: 255.255.248.0.

2. Sequential Allocation (Arranged largest to smallest for natural alignment):

- **Allocation 1 (Org 4 — 8192 IPs):**
 - Range: 198.16.0.0/19 to 198.16.31.255/19
 - Subnet Mask: 255.255.224.0
- **Allocation 2 (Org 1 — 4096 IPs):**
 - Range: 198.16.32.0/20 to 198.16.47.255/20
 - Subnet Mask: 255.255.240.0
- **Allocation 3 (Org 3 — 4096 IPs):**
 - Range: 198.16.48.0/20 to 198.16.63.255/20
 - Subnet Mask: 255.255.240.0
- **Allocation 4 (Org 2 — 2048 IPs):**
 - Range: 198.16.64.0/21 to 198.16.71.255/21
 - Subnet Mask: 255.255.248.0

Total addresses allocated = 8192 + 4096 + 4096 + 2048 = 18 432

<i>Subnet</i>	<i>Network Address</i>	<i>First Usable Host</i>	<i>Last Usable Host</i>	<i>Broadcast Address</i>
Subnet 1	16.14.80.0/22	16.14.80.1	16.14.83.254	16.14.83.255
Subnet 2	16.14.84.0/22	16.14.84.1	16.14.87.254	16.14.87.255
Subnet 3	16.14.88.0/22	16.14.88.1	16.14.91.254	16.14.91.255
Subnet 4	16.14.92.0/22	16.14.92.1	16.14.95.254	16.14.95.255

Unit 3 4 Marks Medium Priority

SEE 2024

Theory Link: Unit 3 § 2

Define Bandwidth-Delay Product (BDP) in networking. What does it physically represent?

Practice Problem:

Model Exam Answer

Solution:

1. Definition and Formula

The **Bandwidth-Delay Product (BDP)** is the product of a link's transmission bandwidth (in bits per second) and its one-way propagation delay (in seconds):

$$\text{BDP} = \text{Bandwidth (bps)} \times \text{Propagation Delay (seconds)}$$

2. Physical Interpretation: 'Volume of the Pipe'

- BDP defines the maximum number of bits that can fill the transmission medium at any given instant.
- Think of the link as a pipe carrying bits. Bandwidth is the cross-sectional area of the pipe, and delay is the length of the pipe. BDP is the total volume of the pipe in bits.
- **Importance:** If a sender wishes to achieve 100% link utilization without stalling, its sender window size must be at least equal to $2 \times \text{BDP}$ (round-trip bandwidth-delay product).

Source: Forouzan 5e, Chapter 3.6, pp. 95–97

Out of CIE-1 Scope Questions (8 Questions)

ARCHIVED

The following questions were identified in past papers but fall outside the official CIE-1 syllabus. They are preserved here to ensure zero questions are dropped.

Paper / Origin	Question Summary	Exclusion Rationale
CIE-2 2025 Q1a	Elaborate on the different error reporting messages and query messages supported by ICMP.	<u>ICMP is in Unit 3 after IPv4 Classless addressing (CIE-1 cutoff).</u>
CIE-2 2025 Q2a, SEE 2024 Q6a, SEE JAN 2023 Q2a	With a neat diagram, discuss the procedure for NAT address translation using one IP address.	<u>NAT is listed after classless addressing in syllabus order.</u>
CIE-2 2025 Q3a, SEE JAN 2026 Q6a, SEE 2025(0) Q3a	Applying Link State Routing, calculate the shortest path from the root node A to all other nodes.	<u>Routing algorithms (Dijkstra, Bellman-Ford) are out of CIE-1 scope.</u>
SEE JAN 2026 Q5a	An IPv4 datagram has arrived with the following header in hex: 45 00 00 ... Calculate length, header length, fragmentation flags.	<u>IPv4 datagram format and fragmentation are out of CIE-1 scope.</u>
CIE-2 2025 Q3b, SEE 2025 Q8b, MAKEUP 2023 Q2b	Discuss the different strategies to achieve transition from IPv4 to IPv6 (Dual stack, Tunneling, Header translation).	<u>IPv6 is part of Unit 4.</u>
SEE 2024 Q10a, SEE 2025(0) Q5a, SEE JAN 2023 Q6a	Explain three-way handshaking in Transmission Control Protocol (TCP) with a neat diagram.	<u>TCP connection management belongs to Unit 5.</u>
SEE 2024 Q9a, SEE 2025 Q7a, SEE 2025 Q9a	Explain the TCP segment structure and how TCP handles lost segments and congestion control.	<u>TCP congestion control is part of Unit 5.</u>

Paper / Origin	Question Summary	Exclusion Rationale
CIE-2 2025 Q3c, SEE 2025 Q10a, MAKEUP 2023 Q7a	Explain how recursive and iterative resolution solve the name-address conflict in Domain Name System (DNS).	<u>DNS is an</u> <u>Application Layer</u> <u>protocol in Unit 5.</u>